

EXAMPLE 1 Bound for Local Truncation Errors

Find a bound for the local truncation errors for Euler's method applied to $y' = 2xy$, $y(1) = 1$.

SOLUTION From the solution $y = e^{x^2-1}$ we get $y'' = (2 + 4x^2)e^{x^2-1}$, so the local truncation error is

$$y''(c) \frac{h^2}{2} = (2 + 4c^2)e^{(c^2-1)} \frac{h^2}{2},$$

where c is between x_n and $x_n + h$. In particular, for $h = 0.1$ we can get an upper bound on the local truncation error for y_1 by replacing c by 1.1:

$$[2 + (4)(1.1)^2]e^{((1.1)^2-1)} \frac{(0.1)^2}{2} = 0.0422.$$

From Table 9.1 we see that the error after the first step is 0.0337, less than the value given by the bound.

Similarly, we can get a bound for the local truncation error for any of the five steps given in Table 9.1 by replacing c by 1.5 (this value of c gives the largest value of $y''(c)$ for any of the steps and may be too generous for the first few steps). Doing this gives

$$[2 + (4)(1.5)^2]e^{((1.5)^2-1)} \frac{(0.1)^2}{2} = 0.1920 \quad (2)$$

as an upper bound for the local truncation error in each step. ■

Note that if h is halved to 0.05 in Example 1, then the error bound is 0.0480, about one-fourth as much as shown in (2). This is expected because the local truncation error for Euler's method is $O(h^2)$.

In the above analysis we assumed that the value of y_n was exact in the calculation of y_{n+1} , but it is not because it contains local truncation errors from previous steps. The total error in y_{n+1} is an accumulation of the errors in each of the previous steps. This total error is called the **global truncation error**. A complete analysis of the global truncation error is beyond the scope of this text, but it can be shown that the global truncation error for Euler's method is $O(h)$.

We expect that, for Euler's method, if the step size is halved the error will be approximately halved as well. This is borne out in Tables 9.1 and 9.2 where the absolute error at $x = 1.50$ with $h = 0.1$ is 0.5625 and with $h = 0.05$ is 0.3171, approximately half as large.

In general it can be shown that if a method for the numerical solution of a differential equation has local truncation error $O(h^{\alpha+1})$, then the global truncation error is $O(h^{\alpha})$.

For the remainder of this section and in the subsequent sections we study methods that give significantly greater accuracy than does Euler's method.

IMPROVED EULER'S METHOD The numerical method defined by the formula

$$y_{n+1} = y_n + h \frac{f(x_n, y_n) + f(x_{n+1}, y_{n+1}^*)}{2}, \quad (3)$$

where

$$y_{n+1}^* = y_n + hf(x_n, y_n), \quad (4)$$

is commonly known as the **improved Euler's method**. To compute y_{n+1} for $n = 0, 1, 2, \dots$ from (3), we must, at each step, first use Euler's method (4) to obtain an initial estimate y_{n+1}^* . For example, with $n = 0$, (4) gives $y_1^* = y_0 + hf(x_0, y_0)$, and then, knowing this value, we use (3) to get $y_1 = y_0 + h \frac{f(x_0, y_0) + f(x_1, y_1^*)}{2}$, where

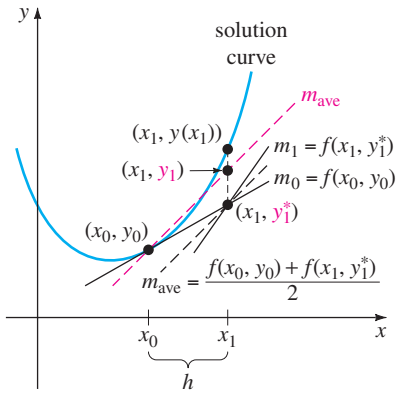


FIGURE 9.1.1 Slope of red dashed line is the average of m_0 and m_1

$x_1 = x_0 + h$. These equations can be readily visualized. In Figure 9.1.1 observe that $m_0 = f(x_0, y_0)$ and $m_1 = f(x_1, y_1^*)$ are slopes of the solid straight lines shown passing through the points (x_0, y_0) and (x_1, y_1^*) , respectively. By taking an average of these slopes, that is, $m_{\text{ave}} = \frac{f(x_0, y_0) + f(x_1, y_1^*)}{2}$, we obtain the slope of the parallel dashed skew lines. With the first step, rather than advancing along the line through (x_0, y_0) with slope $f(x_0, y_0)$ to the point with y -coordinate y_1^* obtained by Euler's method, we advance instead along the red dashed line through (x_0, y_0) with slope m_{ave} until we reach x_1 . It seems plausible from inspection of the figure that y_1 is an improvement over y_1^* .

In general, the improved Euler's method is an example of a **predictor-corrector method**. The value of y_{n+1}^* given by (4) predicts a value of $y(x_n)$, whereas the value of y_{n+1} defined by formula (3) corrects this estimate.

EXAMPLE 2 Improved Euler's Method

Use the improved Euler's method to obtain the approximate value of $y(1.5)$ for the solution of the initial-value problem $y' = 2xy$, $y(1) = 1$. Compare the results for $h = 0.1$ and $h = 0.05$.

SOLUTION With $x_0 = 1$, $y_0 = 1$, $f(x_n, y_n) = 2x_n y_n$, $n = 0$, and $h = 0.1$, we first compute (4):

$$y_1^* = y_0 + (0.1)(2x_0 y_0) = 1 + (0.1)2(1)(1) = 1.2.$$

We use this last value in (3) along with $x_1 = 1 + h = 1 + 0.1 = 1.1$:

$$y_1 = y_0 + (0.1) \frac{2x_0 y_0 + 2x_1 y_1^*}{2} = 1 + (0.1) \frac{2(1)(1) + 2(1.1)(1.2)}{2} = 1.232.$$

The comparative values of the calculations for $h = 0.1$ and $h = 0.05$ are given in Tables 9.3 and 9.4, respectively.

TABLE 9.3 Improved Euler's Method with $h = 0.1$

x_n	y_n	Actual value	Abs. error	% Rel. error
1.00	1.0000	1.0000	0.0000	0.00
1.10	1.2320	1.2337	0.0017	0.14
1.20	1.5479	1.5527	0.0048	0.31
1.30	1.9832	1.9937	0.0106	0.53
1.40	2.5908	2.6117	0.0209	0.80
1.50	3.4509	3.4904	0.0394	1.13

TABLE 9.4 Improved Euler's Method with $h = 0.05$

x_n	y_n	Actual value	Abs. error	% Rel. error
1.00	1.0000	1.0000	0.0000	0.00
1.05	1.1077	1.1079	0.0002	0.02
1.10	1.2332	1.2337	0.0004	0.04
1.15	1.3798	1.3806	0.0008	0.06
1.20	1.5514	1.5527	0.0013	0.08
1.25	1.7531	1.7551	0.0020	0.11
1.30	1.9909	1.9937	0.0029	0.14
1.35	2.2721	2.2762	0.0041	0.18
1.40	2.6060	2.6117	0.0057	0.22
1.45	3.0038	3.0117	0.0079	0.26
1.50	3.4795	3.4904	0.0108	0.31

A brief word of caution is in order here. We cannot compute all the values of y_n^* first and then substitute these values into formula (3). In other words, we cannot use the data in Table 9.1 to help construct the values in Table 9.3. Why not?

TRUNCATION ERRORS FOR THE IMPROVED EULER'S METHOD The local truncation error for the improved Euler's method is $O(h^3)$. The derivation of this result is similar to the derivation of the local truncation error for Euler's method.

Since the local truncation error for the improved Euler's method is $O(h^3)$, the global truncation error is $O(h^2)$. This can be seen in Example 2; when the step size is halved from $h = 0.1$ to $h = 0.05$, the absolute error at $x = 1.50$ is reduced from 0.0394 to 0.0108, a reduction of approximately $(\frac{1}{2})^2 = \frac{1}{4}$.

EXERCISES 9.1

Answers to selected odd-numbered problems begin on page ANS-15.

In Problems 1–10 use the improved Euler's method to obtain a four-decimal approximation of the indicated value. First use $h = 0.1$ and then use $h = 0.05$.

1. $y' = 2x - 3y + 1, y(1) = 5; \quad y(1.5)$
2. $y' = 4x - 2y, y(0) = 2; \quad y(0.5)$
3. $y' = 1 + y^2, y(0) = 0; \quad y(0.5)$
4. $y' = x^2 + y^2, y(0) = 1; \quad y(0.5)$
5. $y' = e^{-y}, y(0) = 0; \quad y(0.5)$
6. $y' = x + y^2, y(0) = 0; \quad y(0.5)$
7. $y' = (x - y)^2, y(0) = 0.5; \quad y(0.5)$
8. $y' = xy + \sqrt{y}, y(0) = 1; \quad y(0.5)$
9. $y' = xy^2 - \frac{y}{x}, y(1) = 1; \quad y(1.5)$
10. $y' = y - y^2, y(0) = 0.5; \quad y(0.5)$
11. Consider the initial-value problem $y' = (x + y - 1)^2, y(0) = 2$. Use the improved Euler's method with $h = 0.1$ and $h = 0.05$ to obtain approximate values of the solution at $x = 0.5$. At each step compare the approximate value with the actual value of the analytic solution.
12. Although it might not be obvious from the differential equation, its solution could “behave badly” near a point x at which we wish to approximate $y(x)$. Numerical procedures may give widely differing results near this point. Let $y(x)$ be the solution of the initial-value problem $y' = x^2 + y^3, y(1) = 1$.
 - (a) Use a numerical solver to graph the solution on the interval $[1, 1.4]$.
 - (b) Using the step size $h = 0.1$, compare the results obtained from Euler's method with the results from the improved Euler's method in the approximation of $y(1.4)$.
13. Consider the initial-value problem $y' = 2y, y(0) = 1$. The analytic solution is $y = e^{2x}$.
 - (a) Approximate $y(0.1)$ using one step and Euler's method.
 - (b) Find a bound for the local truncation error in y_1 .
 - (c) Compare the error in y_1 with your error bound.
 - (d) Approximate $y(0.1)$ using two steps and Euler's method.

- (e) Verify that the global truncation error for Euler's method is $O(h)$ by comparing the errors in parts (a) and (d).

14. Repeat Problem 13 using the improved Euler's method. Its global truncation error is $O(h^2)$.

15. Repeat Problem 13 using the initial-value problem $y' = x - 2y, y(0) = 1$. The analytic solution is

$$y = \frac{1}{2}x - \frac{1}{4} + \frac{5}{4}e^{-2x}.$$

16. Repeat Problem 15 using the improved Euler's method. Its global truncation error is $O(h^2)$.

17. Consider the initial-value problem $y' = 2x - 3y + 1, y(1) = 5$. The analytic solution is

$$y(x) = \frac{1}{9} + \frac{2}{3}x + \frac{38}{9}e^{-3(x-1)}.$$

- (a) Find a formula involving c and h for the local truncation error in the n th step if Euler's method is used.
- (b) Find a bound for the local truncation error in each step if $h = 0.1$ is used to approximate $y(1.5)$.
- (c) Approximate $y(1.5)$ using $h = 0.1$ and $h = 0.05$ with Euler's method. See Problem 1 in Exercises 2.6.
- (d) Calculate the errors in part (c) and verify that the global truncation error of Euler's method is $O(h)$.
18. Repeat Problem 17 using the improved Euler's method, which has a global truncation error $O(h^2)$. See Problem 1. You might need to keep more than four decimal places to see the effect of reducing the order of the error.
19. Repeat Problem 17 for the initial-value problem $y' = e^{-y}, y(0) = 0$. The analytic solution is $y(x) = \ln(x + 1)$. Approximate $y(0.5)$. See Problem 5 in Exercises 2.6.
20. Repeat Problem 19 using the improved Euler's method, which has global truncation error $O(h^2)$. See Problem 5. You might need to keep more than four decimal places to see the effect of reducing the order of error.

Discussion Problems

21. Answer the question “Why not?” that follows the three sentences after Example 2 on page 343.